

# E-Commerce Conversion Prediction

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## Problem Statement

Binary classification (Converted=0/1) from 12 user-level features. Evaluation metric is F1 Score — chosen for its robustness under class imbalance.

## Methodology

- 1. EDA** — Analyzed distributions and class separation. Engagement features (Pages\_Viewed, Time\_On\_Site, Products\_Viewed) showed strongest signal.
- 2. Feature Engineering** — 8 derived features: Pages\_Per\_Minute, Products\_Per\_Page, Engagement\_Score (geometric mean), Income\_Per\_Age, Purchase\_Loyalty (Purchases  $\times$  Discount), High\_Engagement (binary flag), Income\_Band, Age\_Band.
- 3. Models** — XGBoost, LightGBM, and Random Forest trained with 5-Fold Stratified CV. Class imbalance handled via scale\_pos\_weight and class\_weight=balanced.
- 4. Ensemble** — Weighted-average of OOF probabilities (weights  $\propto$  CV F1). Decision threshold swept 0.30–0.70; best OOF F1 = {best\_f1:.4f} at threshold={best\_t:.2f}.

## Key Insights

- **Engagement > Demographics:** Browsing depth is the strongest conversion signal.
- **Discount + Loyalty:** Repeat shoppers shown discounts convert at disproportionately high rates.
- **Threshold tuning:** Lowering threshold from 0.5 to ~0.40 recovers missed converters.

